Addition and multiplication Metrics

Professor

Addition of Matrices :-

If A and B are the two matrices of the same order, then their sum, A+B is defined as the matrix, each element is the sum of the corresponding elements of A and B.

$$A = \begin{bmatrix} 4 & 2 & 5 \\ 1 & 3 & -6 \end{bmatrix}_{2\times 3} \qquad B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 4 \end{bmatrix}_{2\times 3}$$

$$A+B = \begin{bmatrix} 5 & 2 & 7 \\ 4 & 4 & -2 \end{bmatrix}$$

Properties of Matrix Addition

i) Commutative law : $A+B = B+A$

ii) Associative Law.

$$A+(B+C) = (A+B)+C$$

Subtraction of Matrices

The difference of two matrices is obtained by subtracting the elements of the second matrix from the corresponding elements of the first.

$$A - B = \left[a_{ij} - b_{ij} \right]$$

$$A = \begin{bmatrix} 8 & 6 & 4 \\ 1 & 2 & 0 \end{bmatrix}_{2\times3}, \quad B = \begin{bmatrix} 3 & 5 & 1 \\ 7 & 6 & 2 \end{bmatrix}_{2\times3}$$

$$A - B = \begin{bmatrix} 5 & 1 & 3 \\ -6 & -4 & -2 \end{bmatrix}$$

$$A - B = -(B - A)$$

$$B - A = \begin{bmatrix} -5 & -1 & -3 \\ 6 & 4 & 2 \end{bmatrix}$$

Scalar Multiple of Matrix

If a matrix is multiplied by scalar quantity k, then each element is multiplied by k, i.e.

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$3A = \begin{bmatrix} 6 & 9 & 12 \\ 12 & 15 & 18 \\ 21 & 24 & 27 \end{bmatrix}$$

Multiplication

The product of two matrices A and B is only possible if the number of columns in A is equal to the number of rows in B

$A \times B = C$ $\quad C_{ij} = a_{i1} b_{1j} + a_{i2} b_{2j} + a_{i3} b_{3j} + \cdots + a_{in} b_{nj}$

$a_{ij} \; b_{ij} = c_{ij}$ $\quad C_{11} = a_{11} b_{11} + a_{12} b_{21} + a_{13} b_{31} + \cdots + a_{1n} b_{n1}$

Q $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}_{3\times 3}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 0 \\ 2 & -1 \end{bmatrix}_{3\times 2}$

$$A\times B = \begin{bmatrix} 0\times 1 + 1\times -1 + 2\times 2 & 0\times -2 + 1\times 0 + 2\times -1 \\ 1\times 1 + 2\times -1 + 3\times 2 & 1\times -2 + 2\times 0 + 3\times -1 \\ 2\times 1 + 3\times -1 + 4\times 2 & 2\times -2 + 3\times 0 + 4\times -1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 3 & -2 \\ 5 & -5 \\ 7 & -8 \end{bmatrix}_{3\times 2}$$

$B \times A = BA$

$$B = \begin{bmatrix} 1 & -2 \\ -1 & 0 \\ 2 & -1 \end{bmatrix}_{3\times \underline{\underline{2}}} \qquad A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}_{\underline{\underline{3}}\times 3}$$

$$2 \neq 3$$

BA is not defined

Q. If $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}$

find the products AB and BA, also show $AB \neq BA$

$$\boxed{(AB)' = B'A'}$$

Properties of Matrix Multiplication

i) Multiplication of matrices is not commutative

$$AB \neq BA$$

ii) Associative $\quad A(BC) = (AB)C$.

iii) Distributive <u>wrt to addition</u>

$$A(B+C) = AB + AC.$$

iv) Multiplication by Identity matrix

$$IA = A$$

v) Multiplication inverse of a matrix exists if $|A| \neq 0$.

$$A \cdot A^{-1} = A^{-1} A = I$$

Q. If $A = \begin{bmatrix} 1 & 2 \\ -2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} -3 & 1 \\ 2 & 0 \end{bmatrix}$

Verify That

$$(AB)C = A(BC)$$

and $A(B+C) = AB + A$